Adilkhan Salkimbayev

Applied Mathematics & Scientific Machine Learning Researcher
+77078261934 | adilsalkimbayev@gmail.com | Almaty, Kazakhstan
hecatetdm.tech
GitHub: github.com/lArtemis13l

EDUCATION

Kazakh-British Technical University (KBTU)
B.Sc. in Automation and Control | GPA: 3.58 | Aug 2023 – July 2027 (Expected)

- **Relevant Coursework:** Linear Algebra for Engineers, Differential Equations, Calculus of Complex Variables
- **Military Department:** Sergeant of the Reserve (Completed June 2025).

PUBLICATIONS AND PREPRINTS

- **A. Salkimbayev.** "Symbolic Kolmogorov-Arnold Networks as a Constant-Time Alternative to Linear Time-Varying Model Predictive Control for Embedded Control of Non-Minimum Phase Systems". *Under Review, Engineering Applications of Artificial Intelligence.*
- **A. Salkimbayev, K.S. Haider.** "Robust Adaptive Kolmogorov-Arnold Neural Control." *In Preparation for Automatica.*

RESEARCH EXPERIENCE

Independent Undergraduate Researcher | Kazakh-British Technical University

Functional-Analytic Control Architecture & PDEs (Current)
- Developing a novel functional-analytic control architecture drawing upon operator theory and methodologies from Brezis, Evans and Stein.
- Translating control frameworks into rigorous mathematical formulations applicable to the study of partial differential equations and fluid dynamics.

Adaptive Control & Nonlinear Dynamical Systems (Jan 2026 – Jul 2026)
- Developed a learning-based adaptive control architecture combining Recursive Least Squares with symbolic Kolmogorov-Arnold Networks (KANs).
- Derived rigorous Lyapunov stability guarantees (ISS / GUUB) utilizing advanced stability arguments for nonlinear system regulation.
- Evaluated algorithm on a quadruple-tank benchmark system via Python simulations, demonstrating a deterministic computational runtime and consistent reduction in the Total Variation of the control input.

Constant-Time Neural Control for Embedded Systems (Nov 2025 – Apr 2026)
- Designed a symbolic KAN architecture as a computationally constrained alternative to Linear Time-Varying Model Predictive Control (MPC) for non-minimum phase systems.
- Executed Hardware-in-the-Loop (HIL) methodology on an STM32H7 microcontroller, achieving a runtime reduction of up to 5 orders of magnitude (OOM).

ADVANCED INDEPENDENT STUDY

- **Functional Analysis, Sobolev Spaces and Partial Differential Equations** (Haim Brezis)
- **Partial Differential Equations** (Lawrence C. Evans)
- **Real Analysis: Measure Theory, Integration, and Hilbert Spaces** (Elias M. Stein & Rami Shakarchi)
- **Singular Integrals and Differentiability Properties of Functions** (Elias M. Stein)

- **Introduction to Real Analysis** (William F. Trench)

TECHNICAL & COMPUTATIONAL SKILLS

Mathematics & Control Systems:
Operator theory, partial differential equations, semigroup theory, Lyapunov stability analysis, adaptive control, nonlinear control systems

Scientific Machine Learning:
Kolmogorov-Arnold Networks, system identification, embedded SciML.

Programming & Hardware: Python (NumPy, SciPy, matplotlib, pandas, PyTorch), C/C++, STM32 CubeIDE, LaTeX (VSCode).